24/12/2020

Aakash
Medical | IIT-JEE | Foundations
(Divisions of Aakash Educational Services Limited)
Regd. Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph. 011-47623456

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 2

Topics covered :

Physics : Laws of Motion, Work, Energy & Power

Chemistry : Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, States of Matter : Gases and Liquids

Botany : Plant Kingdom, Morphology of Flowering Plants

Zoology : Structural Organisation in Animals, Biomolecules

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. Two balls of mass 1 kg and 2 kg are thrown upward simultaneously with the same speed. If the air exerts a constant and equal force of resistance on the two balls, then maximum heights covered by 1 kg and 2 kg are h_1 and h_2. The correct relation between h_1 and h_2 is

(1) $h_1 > h_2$ (2) $h_2 > h_1$

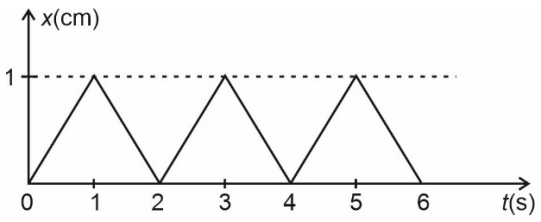
(3) $h_1 = h_2$ (4) $h_1 = 2h_2$

2. A stone of mass m tied to the end of a string revolves in a vertical circle of radius r. If the speed of the stone at the lowest point is v, then the net force on stone at the lowest point directed vertically downwards is (T represents the tension in the string at the lowest point)

(1) $mg + T$ (2) $mg + T + \frac{mv^2}{r}$

(3) $mg - T$ (4) $mg - T - \frac{mv^2}{r}$

3. The position-time graph of a ball of mass 0.05 kg is given in the diagram. The magnitude of net impulse received by the ball just after five seconds is



(1) 10^{-3} kg m/s (2) Zero

(3) 5×10^{-3} kg m/s (4) 1.5×10^{-3} kg m/s

4. A stream of water flowing horizontally with a speed of 10 m s^{-1} gushes out of a tube of cross-sectional area 10 cm^2, and hits a vertical wall nearby. If water rebounds with same speed, then the force exerted on the wall by the impact of water is

(1) 1250 N (2) 2250 N

(3) 100 N (4) 200 N

5. A man stands in contact against the inner wall of a hollow cylindrical drum of radius *R* rotating about its vertical axis with some angular speed. If the coefficient of friction between the wall and his clothing is 0.5 and floor is suddenly removed then the possible rotational speed of the cylinder to enable the man to remain stuck to the wall (without falling) is

(1) $\sqrt{\frac{g}{R}}$ (2) $\sqrt{\frac{g}{2R}}$

(3) $2\sqrt{\frac{g}{R}}$ (4) $\sqrt{\frac{g}{4R}}$

6. A block of mass 12 kg is placed on a long trolley. The coefficient of static friction between the block and the trolley is 0.2. If trolley accelerates from rest with 0.2 m s^{-2}, then the net force on the block is

(1) 24 N (2) Zero

(3) 2.4 N (4) 21.6 N

7. A bob of mass 0.2 kg hung from the ceiling of a room by a string 1 m long is set into oscillation. The speed of the bob at its mean position is 2 m s^{-1}. If the string is cut when the bob is at a point between its extreme and mean position, then the trajectory of the bob is

(1) Straight line

(2) Parabola

(3) Circle

(4) Ellipse

8. An aircraft execute a horizontal loop at a speed of 360 km/h with its wings banked at 37°. The radius of the loop is nearly

(1) 13 km

(2) 15 km

(3) 4.5 km

(4) 1.3 km

9. A circular racetrack of radius 120 m is banked at an angle of 45°. If the coefficient of friction between the wheels of a race-car and the road is 0.5 then the maximum permissible speed to avoid slipping is

(1) 60 km/h (2) 200 km/h

(3) 68 km/h (4) 216 km/h

10. A rocket with a lift-off mass 15,000 kg is blasted upwards with an initial acceleration of 4 m/s^2. The initial thrust (force) of the blast is

(1) Zero (2) 6×10^4 N

(3) 2.1×10^5 N (4) 1.5×10^5 N

11. Two billiard balls each of mass 0.04 kg moving in opposite directions with speed 8 m/s collide and rebound with the same speed. The change in the magnitude of momentum of each ball is

(1) 0.32 kg m s^{-1} (2) 0.64 kg m s^{-1}

(3) Zero (4) 1.28 kg m s^{-1}

12. A block of mass *m* is kept on a rough horizontal surface having coefficient of friction of 0.5. A horizontal force is applied on the block, but the block remains at rest. The maximum contact force exerted by the surface on the block can be

(1) $\frac{mg}{2}$ (2) $\frac{\sqrt{5}mg}{2}$

(3) $\frac{3mg}{2}$ (4) mg

13. A disc revolves with a speed of $\frac{5}{3}$ rev/s and has a radius of 10 cm. If a coin is placed at 9 cm away from the centre of the disc, the coin just slips. The coefficient of friction is (Take g = 10 m/s^2 and π^2 = 9.8)

(1) 0.1 (2) 0.49

(3) 0.98 (4) 0.2

14. The rear side of a truck is open and a block of mass 10 kg is placed 6 m away from the open end. The coefficient of friction between the block and the surface is 0.1. On a straight road, if truck starts from rest and accelerates with 4 m/s^2, then time taken by the block to fall off the truck is

(1) $\sqrt{2}$ s (2) 4 s

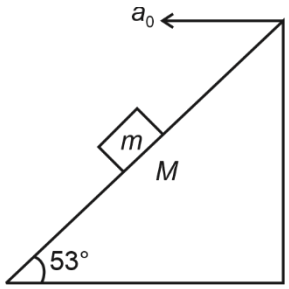
(3) 2 s (4) 6 s

15. A bomb explodes into three fragments in a horizontal plane. If first part of 1 kg moves with velocity $(\hat{i}+2\hat{j})\,\text{m s}^{-1}$ and second part of mass 2 kg is moving with velocity $(2\hat{i}-3\hat{j})\,\text{m s}^{-1}$ then the velocity of third part of mass 1 kg is

(1) $(4\hat{i}-5\hat{j})\,\text{m s}^{-1}$ (2) $(4\hat{i}+5\hat{j})\,\text{m s}^{-1}$

(3) $(5\hat{i}-4\hat{j})\,\text{m s}^{-1}$ (4) $(-5\hat{i}+4\hat{j})\,\text{m s}^{-1}$

16. A block of mass *m* is placed on the wedge of mass *M* which is moving with constant horizontal acceleration a_0. If all the surfaces are smooth and block is at rest with respect to wedge then normal reaction on the block by the wedge will be



(1) $\frac{m}{5}[4g+3a_0]$ (2) $\frac{m}{5}[4g-3a_0]$

(3) $\frac{m}{5}[3g+4a_0]$ (4) $\frac{m}{5}[3g-4a_0]$

17. A car of mass *m* is moving on a horizontal circular track. If coefficient of static friction between the road and the tyres of the car is 0.75, then the maximum contact force acted by the road on the car is

(1) $\frac{mg}{4}$ (2) $\frac{5mg}{4}$

(3) $\frac{4mg}{5}$ (4) $\frac{7mg}{4}$

18. A light string passing over a smooth light pulley connected two blocks of different masses in vertical direction. If the acceleration of the block is $\frac{g}{8}$, then the ratio of the masses is

(1) $\frac{9}{7}$ (2) $\frac{6}{5}$

(3) 3 (4) $\frac{5}{2}$

19. A person of mass 30 kg is inside a lift of mass 970 kg and presses the button on control panel. The lift starts moving downward with an acceleration a_0. If tension in the supporting cable is 8000 N, then the value of a_0 is

(1) 2 m/s^2 (2) 12 m/s^2

(3) 0.5 m/s^2 (4) 1 m/s^2

20. If net external force acting on a body is $\vec{F}$ to produce an acceleration $\vec{a}$, then select correct option

(1) $\vec{F}\cdot\vec{a}=0$

(2) $\vec{F}\times\vec{a}=0$

(3) $\vec{F}\cdot\vec{a}>0$

(4) Both (2) and (3) are correct

21. A rain drop of mass *m* is falling downward with a constant speed. The net force acting on it is

(1) *mg* (2) Zero

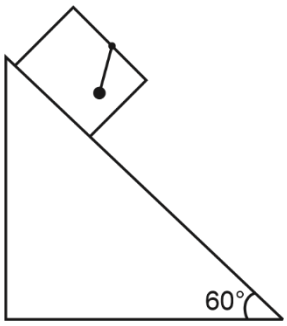
(3) Infinite (4) $\frac{mg}{2}$

22. The motion of a particle of mass *m* is described by $y = ut + gt^2$. The force acting on the particle is [where *g* is the acceleration due to gravity]

(1) *mg* (2) Zero

(3) $\frac{mg}{2}$ (4) 2*mg*

23. A pendulum is hanging from the ceiling of a cart which is moving down a smooth inclined plane as given in the figure. The angle made by string with normal to the ceiling in equilibrium is



(1) 0°

(2) 60°

(3) 90°

(4) 120°

24. A ball of mass m strikes a horizontal surface with speed v at an angle 90° with surface. It gets reflected with speed $\frac{v}{2}$ at an angle 90° to the surface. The magnitude of impulse imparted by the surface to the ball is

(1) $2mv$ (2) $\frac{mv}{2}$

(3) $\frac{3mv}{2}$ (4) mv

25. Two particles of mass m *and* $3m$ are whirled in horizontal circles of radius r and $\frac{r}{2}$ respectively. If tangential speed of lighter particle is two times that of the value of heavier particle, then the ratio of centripetal force acting on the particles is

(1) $\frac{2}{3}$ (2) $\frac{4}{3}$

(3) $\frac{8}{3}$ (4) $\frac{1}{24}$

26. The projection of force $\vec{F} = (2\hat{i} - \hat{j} + 2\hat{k})$ N on the displacement $\vec{d} = (\hat{i} + \hat{j} + \hat{k})$ m is

(1) $\sqrt{3}$ J (2) $\frac{1}{\sqrt{3}}$ J

(3) $\frac{1}{\sqrt{3}}$ N (4) $\sqrt{3}$ N

27. A raindrop of volume 1 ml is falling from a height 1 km. If it hits the ground with a speed of 180 km h^{-1}, then the work done by the resistive force is

(1) 8.75 J

(2) 1.25 J

(3) –8.75 J

(4) –1.25 J

28. A cyclist comes to a skidding stop in 8 m. During this process, the force on the cycle due to the road is 150 N and is directly opposed to the motion. The work done by the cycle on the road is

(1) 1200 J

(2) Zero

(3) 120 J

(4) –1200 J

29. A woman pushes a trunk on a railway platform which has a rough surface. She applies a force of 50 N over a distance of 12 m. Thereafter, she gets progressively tired and her applied force reduces linearly with distance to 25 N. If the total distance through which the trunk has been moved is 24 m, then the work done by the woman over 24 m is

(1) 1750 J (2) 150 J

(3) 1050 J (4) –1050 J

30. Two forces of 6 N and 2 N are acting on a particle along x-axis and y-axis simultaneously. The net work done to displace the particle from (0, 0) to (3, 4) m is

(1) $10\sqrt{10}$ J (2) 26 J

(3) 18 J (4) 40 J

31. A particle of mass 1 kg starting from rest undergoes uniform acceleration. If the speed acquired in time 2 s is 4 m/s, then power at t = 1 s is

(1) 8 W (2) 4 W

(3) 2 W (4) 16 W

32. A bob of mass m is suspended by a light string of length L. It is imparted a horizontal velocity v at the lowest point such that it completes a semi-circular trajectory in the vertical plane with the string becoming slack only on reaching the topmost point. The ratio of kinetic energy at lowest and topmost point is

(1) 6 (2) 3

(3) 5 (4) 2

33. A bullet of mass 0.02 kg and horizontal speed 41 m s^{-1} strikes a block of wood of mass 0.8 kg and instantly comes to rest with respect to the block. The block is suspended from the ceiling by means of thin wires. The height to which the block rises, is

(1) 0.5 cm (2) 5 cm

(3) 0.4 m (4) 0.5 m

34. A bolt of mass 0.2 kg falls from the ceiling of an elevator moving down with an uniform speed of 5 m/s. It hits the floor of the elevator (length of the elevator is 4 m) and does not rebound. The heat produced by the impact is (g = 9.8 m/s^2)

(1) 7.81 J (2) 7.84 J

(3) Zero (4) 7.74 J

35. A body of mass 2 kg travels in a straight line with velocity $v = x^{3/2}$. The work done by the net force during its displacement from $x = 0$ to $x = 4$ m is

(1) 32 J (2) 64 J

(3) 16 J (4) 50 J

36. An object of mass 4 kg is lying in *x*-*y* plane at rest. It suddenly explodes into three pieces. Two pieces, each of mass 1 kg move perpendicular to each other with equal speeds of 2 m/s. The total kinetic energy generated due to explosion is

(1) 10 J (2) 8 J

(3) 12 J (4) 6 J

37. A particle of mass 2 kg begins to move under the action of a time dependent force $\vec{F} = (t\hat{i} + 2\hat{j})$ N. The power developed by the force at $t = 1$ s, is

(1) 4.50 W (2) 8.75 W

(3) 6.75 W (4) 2.25 W

38. A small block of mass 0.01 kg is whirled in a vertical circle by attaching it to a light stick of length 1 m. The minimum speed of the block at horizontal orientation of stick to complete the circle is

(1) $\sqrt{g}$ (2) $\sqrt{2g}$

(3) $\sqrt{4g}$ (4) $\sqrt{3g}$

39. Two identical balls each of mass 2 kg are moving towards each other with speeds of 0.4 m/s and 0.2 m/s respectively collide elastically in one dimension. The impulse imparted by one ball on other is

(1) 1.2 N s

(2) 0.4 N s

(3) 0.8 N s

(4) Zero

40. A moving ball having mass of 0.05 kg, collides with another stationary ball having mass of 250 g. The lighter ball comes to rest after collision. If the initial velocity of the lighter ball is 6 cm/s, then the value of coefficient of restitution (*e*) will be

(1) 0.5

(2) 0.4

(3) 0.25

(4) 0.2

41. A body of mass 1 kg moving with speed 2 m/s collides elastically with another body of mass 1.5 kg initially at rest. The fractional kinetic energy retained by the lighter body is

(1) $\frac{3}{10}$ (2) $\frac{1}{24}$

(3) $\frac{1}{25}$ (4) $\frac{24}{25}$

42. The potential energy of a particle in a force field is $U = \frac{\alpha}{r^2} - \frac{\beta}{r}$, where α and β are positive constants and *r* is the distance of particle from the centre of the field. For stable equilibrium, the minimum potential energy is

(1) $-\frac{\beta^2}{4\alpha}$ (2) $\frac{\beta^2}{2\alpha}$

(3) $-\frac{\beta^2}{2\alpha}$ (4) $\frac{\beta^2}{4\alpha}$

43. A particle is dropped from a height *H*. It hits the ground with a certain kinetic energy *K*. If the same particle is dropped from a height 200% more than the previous height, the kinetic energy when it hits the ground will change by

(1) 41% (2) 17.3%

(3) 200% (4) 100%

44. A stone is projected with a velocity $\vec{u} = (3\hat{i} + 4\hat{j})$ m/s from the ground in a vertical plane under the effect of gravity. The power exerted by the gravitational force is

(1) Zero at the instant just after, the stone is projected

(2) Zero at the highest position of the stone

(3) Zero at the instant just before, the stone hits the ground

(4) Never zero during its motion.

45. A car starts from rest and accelerates so that the instantaneous power delivered to the car is constant. If the instantaneous velocity of the car at $t = 1$ s is *v*, then the velocity at $t = 8$ s is

(1) $2v$ (2) $2\sqrt{2}\,v$

(3) $\frac{v}{2\sqrt{2}}$ (4) $\frac{v}{2}$

CHEMISTRY

46. In which of the following options, the order of arrangement does not agree with the variation of property indicated against it?
(1) F > O > N > C (Electronegativity)
(2) Si > P > S > Cl (Atomic radii)
(3) Rb > K > Na > Li (Metallic character)
(4) O > N > C > B (Ionisation enthalpy)

47. The element which has highest positive electron gain enthalpy is
(1) Be (2) N
(3) He (4) Ne

48. Most acidic compound among the following is
(1) CO_2 (2) Cl_2O_7
(3) SO_3 (4) P_4O_{10}

49. Ion which has the largest size is
(1) Al^{3+} (2) K^+
(3) S^{2-} (4) Ca^{2+}

50. The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$. What is the atomic number of the element, which is just below the above given element in the periodic table?
(1) 35 (2) 56
(3) 50 (4) 49

51. The atom which has electronic configuration, $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$ belongs to
(1) Period 4, group 16
(2) Period 3, group 14
(3) Period 3, group 16
(4) Period 4, group 14

52. IUPAC official name of element with atomic number 107 is
(1) Mendelevium (2) Seaborgium
(3) Dubnium (4) Bohrium

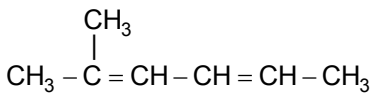
53. The general electronic configuration of *f*-block elements is
(1) $(n-2)\ f^{1\text{-}14}\ nd^{0-1}\ ns^2$
(2) $(n-2)\ f^{1\text{-}14}\ (n-1)\ d^{0-1}\ ns^2$
(3) $(n-2)\ f^{1\text{-}14}\ (n-1)\ d^{10}\ ns^2$
(4) $(n-2)\ f^{1\text{-}14}\ (n+1)\ d^2\ ns^2$

54. Maximum number of elements present in the fifth period of modern periodic table is
(1) 8 (2) 10
(3) 18 (4) 32

55. The pair of species which are isostructural is
(1) CCl_4, XeF_4 (2) I_3^+, C_2H_2
(3) NH_3, BF_3 (4) NO_3^-, SO_3

56. The compound which contains both polar and non-polar bonds is
(1) H_2O_2 (2) PF_2Cl_3
(3) SF_6 (4) XeF_4

57. The total number of σ bonds and π electrons present in the following compound respectively are



(1) 18, 4 (2) 18, 2
(3) 16, 2 (4) 20, 4

58. According to MOT, the species which has only π bond(s) is
(1) B_2 (2) H_2
(3) Li_2 (4) F_2

59. Non-polar molecule among the following is
(1) H_2O (2) CO_2
(3) SO_2 (4) NO_2

60. The number of electrons present in antibonding molecular orbitals in N_2^+ ion is
(1) 5 (2) 4
(3) 6 (4) 7

61. Species which has highest bond order is
(1) O_2^+ (2) CN^-
(3) N_2^- (4) C_2

62. Hybridisation of Xe in XeO_3, $XeOF_4$ and XeF_6 respectively are
(1) sp^3, sp^3d, sp^3d^3 (2) dsp^2, sp^3d, sp^3d^3
(3) dsp^2, sp^3d^3, sp^3d^2 (4) sp^3, sp^3d^2, sp^3d^3

63. Species which has square planar shape is

(1) $[PtCl_4]^{2-}$ (2) XeO_4

(3) SF_4 (4) CH_4

64. Total number of lone-pairs of electrons present in ClF_3 is

(1) 11 (2) 12

(3) 13 (4) 10

65. Incorrect statement about PCl_5 molecule is

(1) Hybridisation of P is sp^3d

(2) Shape of molecule is trigonal bipyramidal

(3) All the bond angles are not equivalent

(4) Three bonds are longer than the other two bonds

66. Which among the following hybrid orbitals has the maximum % s-character?

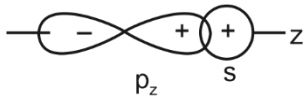
(1) *sp*

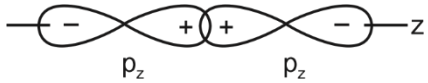
(2) sp^2

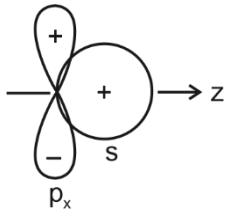
(3) sp^3

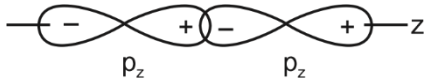
(4) sp^3d

67. Which of the following represents negative overlapping?

(1) 

(2) 

(3) 

(4) 

68. Correct order of bond length is

(1) C–N > O–H > C–H

(2) C–H > C–N > O–H

(3) C–N > C–H > O–H

(4) O–H > C–H > C–N

69. Maximum covalent character is present in

(1) NaF (2) LiF

(3) KF (4) CsF

70. Identify the incorrect match among the following.

(1) O_2 : Paramagnetic

(2) N_2 : Diamagnetic

(3) O_2^{2-} : Diamagnetic

(4) NO : Diamagnetic

71. Correct order of bond angle is

(1) $\overset{\oplus}{N}H_4 > NH_3 > \overset{\ominus}{N}H_2$

(2) $\overset{\ominus}{N}H_2 > NH_3 > \overset{\oplus}{N}H_4$

(3) $NH_3 > \overset{\ominus}{N}H_2 > \overset{\oplus}{N}H_4$

(4) $\overset{\oplus}{N}H_4 > \overset{\ominus}{N}H_2 > NH_3$

72. Intramolecular H-bonding exist in

(1) HF (2) NH_3

(3) o-nitrophenol (4) p-nitrophenol

73. C–O bond order in CO_3^{2-} ion is

(1) 1.75 (2) 1.25

(3) 2 (4) 1.33

74. Equal masses of H_2 and He are taken in a container at 25°C. The fraction of total pressure exerted by H_2 is

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$

(3) $\frac{2}{3}$ (4) $\frac{1}{3}$

75. 1 atm pressure is not equivalent to

(1) 760 torr (2) 76 mm Hg

(3) 101325 pascals (4) 1.013 bar

76. An open flask containing air is heated from 250 K to 350 K. The percentage of air escaped from the container is

(1) 40.1% (2) 21.2%

(3) 28.6% (4) 39.2%

77. The temperature at which the average speed of SO_2 is same as that of O_2 at 27°C is

(1) 750 K

(2) 500 K

(3) 450 K

(4) 600 K

78. Same volume of an unknown gas X and CH_4 diffuse through a porous partition in 40 and 20 seconds respectively. The molar mass of the unknown gas X is

(1) 32 u (2) 64 u

(3) 4 u (4) 68 u

79. Maximum deviation of a gas from ideal behaviour is observed at

(1) High temperature and high pressure

(2) High temperature and low pressure

(3) Low temperature and high pressure

(4) Low temperature and low pressure

80. A gas at a certain temperature and pressure has molar volume 5 percent greater than that of an ideal gas under the same conditions. The correct option among the following is

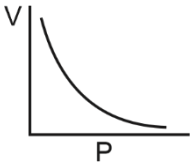
(1) Z > 1, attractive forces are dominant

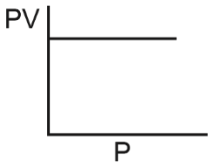
(2) Z > 1, repulsive forces are dominant

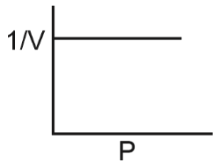
(3) Z < 1, attractive forces are dominant

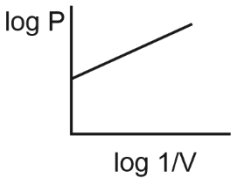
(4) Z < 1, repulsive forces are dominant

81. Which of the following graph is incorrect for Boyle's law?

(1) 

(2) 

(3) 

(4) 

82. The correct order of van der Waals constant 'a' for the following gases is

(1) He > O_2 > CO > CO_2

(2) CO > CO_2 > O_2 > He

(3) CO_2 > CO > O_2 > He

(4) CO_2 > O_2 > CO > He

83. The density of CO_2 gas at 27°C and 1 atm pressure is

(1) 0.98 g/L (2) 2.5 g/L

(3) 1.79 g/L (4) 3.6 g/L

84. Correct relation between average molar kinetic energy of NO and N_2O at same temperature is

(1) $KE_{(NO)} = 2KE_{(N_2O)}$

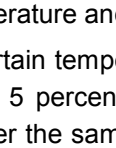
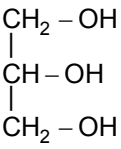
(2) $KE_{(NO)} = \frac{2}{3}KE_{(N_2O)}$

(3) $KE_{(NO)} = KE_{(N_2O)}$

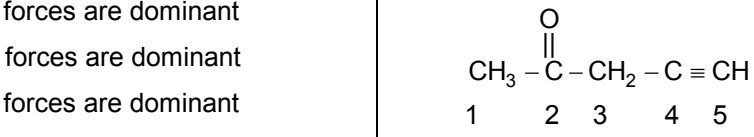
(4) $KE_{(NO)} = \frac{3}{2}KE_{(N_2O)}$

85. Surface tension is maximum for

(1) C_2H_5OH (2) H_2O

(3)  (4) 

86. The hybridisation of C_1, C_2 and C_5 in the following compound respectively are



(1) sp^3, sp^2, sp (2) sp^3, sp, sp^2

(3) sp, sp^2, sp^3 (4) sp^2, sp^2, sp

87. Lithium shows diagonal relationship with

(1) Na (2) Mg

(3) Al (4) Si

88. Shape of ICl_2^- ion is

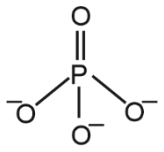
(1) Bent (2) Linear

(3) Trigonal planar (4) T-shape

89. Which of the following represents an incorrect relation?

(1) V_C = 3b (2) $T_C = \frac{a}{27\,Rb}$

(3) $P_C = \frac{a}{27\,b^2}$ (4) $T_i = \frac{2a}{Rb}$

90. Formal charge on P in  is

(1) +1 (2) +2

(3) +5 (4) Zero

BOTANY

91. Select the **wrongly** matched option.
(1) Red algae – Flagellated gametes
(2) *Marchantia* – Dioecious
(3) *Pinus* – Branched stem
(4) *Riccia* – Liverwort

92. Heterospory is seen in
(1) *Funaria* (2) *Sphagnum*
(3) *Selaginella* (4) *Dryopteris*

93. Which of the following statement is **incorrect**?
(1) Ovules are naked in gymnosperms
(2) Winged pollen grains are seen in *Pinus*
(3) *Cycas* is dioecious
(4) The dominant, main plant body in *Salvinia* is gametophyte

94. Which of the following is a colonial alga?
(1) *Fucus* (2) *Ulothrix*
(3) *Volvox* (4) *Chlorella*

95. Most of the algae show haplontic life cycle, but *Ectocarpus* shows
(1) Haplontic life cycle pattern
(2) Diplontic life cycle pattern
(3) Haplodiplontic life cycle pattern
(4) Usually haplontic but sometimes diplontic pattern

96. All of the following are true for leaves of gymnosperms, **except**
(1) Have thick cuticle
(2) Lack stomata
(3) Can be needle like
(4) Are well adapted to withstand extremes of temperature

97. Select the **wrong** statement.
(1) Algin is a hydrocolloid obtained from brown algae
(2) Atleast half of the total CO_2 fixation on earth is carried out by algae through photosynthesis
(3) *Chlorella* is rich source of protein
(4) Agar is obtained from *Gelidium* and *Sargassum*

98. One of the tallest gymnosperms is
(1) *Pinus* (2) *Cycas*
(3) *Cedrus* (4) *Sequoia*

99. Gametophyte does **not** have free living independent existence in
(1) *Marchantia* (2) *Funaria*
(3) *Pinus* (4) *Sphagnum*

100. Which of the following plant groups have homosporous as well as heterosporus species?
(1) Bryophytes (2) Pteriodophytes
(3) Gymnosperms (4) Angiosperms

101. Select the plant which has archegonia and produce seeds also.
(1) *Funaria* (2) *Selaginella*
(3) *Cycas* (4) Mango

102. Select the **correct** statement.
(1) Brown algae has chlorophyll *a* and *b*.
(2) Coralloid roots of *Cycas* have N_2 fixing cyanobacteria
(3) Triploid endosperm is characteristic feature of gymnosperms
(4) *Marsilea* is homosporous

103. Peat is good manure, ____ is responsible for peat formation.
(1) *Sphagnum* (2) *Funaria*
(3) *Riccia* (4) *Selaginella*

104. Isogamy with non flagellated gametes is seen in
(1) *Volvox* (2) *Fucus*
(3) *Ectocarpus* (4) *Spirogyra*

105. Select the **incorrect** statement.
(1) Mannitol is stored food in red algae
(2) The most common spore produced in algae are zoospores
(3) Pyrenoids contain proteins and starch
(4) *Chlamydomonas* is a green alga

106. *Cycas* differs from *Adiantum* in having / being
(1) Vascular tissues (2) Seeds
(3) Homosporous (4) Sporophyte

107. Select the **incorrect** statement about diplontic life cycle.
(1) Haploid gametophyte is short lived
(2) Shown by *Fucus*
(3) Dominant phase is non-photosynthetic sporophytic plant
(4) Seed bearing plants also show this cycle

108. The seed plants with ovary are
(1) Angiosperms
(2) Gymnosperms
(3) Bryophytes
(4) Pteridophytes

109. Endosperm in flowering plants is formed by A and is generally B .

	A	B
(1)	Syngamy	Diploid
(2)	Triple fusion	Triploid
(3)	Syngamy	Triploid
(4)	Triple fusion	Diploid

110. Spread of pteridophytes is restricted to narrow geographical regions due to
(1) Heterospory
(2) Need of water for fertilisation
(3) Requirement of cool, damp and shady places by gametophytes
(4) Both (2) and (3)

111. Read the following statements and choose the **correct** option.
Statement A : Phylogenetic classification system is based on evolutionary relationships between the various organisms.
Statement B : Cytotaxonomy is based on chromosome number, structure, behaviour etc.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

112. Embryophytes which do not have archegonia are
(1) Mosses (2) Liverworts
(3) Flowering plants (4) Ferns

113. The event unique to angiosperms is
(1) Heterospory
(2) Syngamy
(3) Double fertilisation
(4) Pollen grain formation

114. Modified edible adventitious root is
(1) Radish (2) Turnip
(3) Carrot (4) Sweet potato

115. In root, root hair develop from the zone of
(1) Elongation
(2) Maturation
(3) Root cap
(4) Meristematic activity

116. Select the **incorrect** one about pneumatophores
(1) Seen in *Rhizophora*
(2) Are respiratory roots
(3) Positively geotropic
(4) Have lenticels

117. Mark the common feature of both coconut and mango fruit.
(1) Both have edible endosperm
(2) The mesocarp is fleshy and juicy
(3) Both are drupe
(4) The epicarp is fibrous

118. Leaves modify into small, sharp-pointed structures called leaf spines as in
(1) Sweet pea
(2) *Bougainvillea*
(3) Cactus
(4) Cucumber

119. Axile placentation is seen in
(1) Pea
(2) Lemon
(3) Mustard
(4) Sunflower

120. Radial symmetry is found in the flowers of
(1) Pea
(2) Gulmohur
(3) Chilli
(4) *Cassia*

121. Perigynous flowers are seen in
(1) Guava (2) Mustard
(3) Plum (4) China rose

122. If margin of petals or sepals overlap each other but not in a particular direction it is called _____ aestivation.
(1) Twisted (2) Imbricate
(3) Valvate (4) Vexillary

123. Staminode is
(1) Anther with two pollen sacs filled with pollens
(2) Stamen containing fused microsporangia
(3) Fusion of stamen with other floral parts
(4) Sterile stamen

124. Epiphyllous condition is A as seen in B. Fill the blanks A and B with appropriate option.

	A	B
(1)	Attachment of stamen to the perianth	Brinjal
(2)	Attachment of stamen to the perianth	Lily
(3)	Attachment of stamen to the sepals	Lily
(4)	Attachment of stamen to the sepals	Brinjal

125. Parthenocarpic fruits
(1) Have seeds without seed coat
(2) Are also known as pseudocarpic fruits
(3) Are formed without fertilisation of ovary
(4) Are fertilised ripened ovary

126. Select the **incorrect** match.

(1)	Calyx	–	Generally green in colour
(2)	Unisexual flower	–	China rose
(3)	Perianth	–	Tepal
(4)	Bilateral symmetry	–	Bean

127. Syncarpous condition is seen in
(1) Lotus and mustard (2) Rose and tomato
(3) Mustard and tomato (4) Lotus and rose

128. Select the **incorrect** statement about racemose inflorescence.
(1) Flowers are borne in acropetal succession
(2) Main axis continues to grow
(3) Younger flowers are present at the base
(4) Exemplified by mustard

129. Single leaf is present at each node in an alternate fashion in
(1) Guava (2) *Calotropis*
(3) Mustard (4) *Alstonia*

130. Potato tuber is modified
(1) Leaf
(2) Adventitious root
(3) Stem
(4) Tap root

131. Mark the **incorrect** option about phylloclade.
(1) It is photosynthetic
(2) Has unlimited growth
(3) Seen in *Opuntia*
(4) It is underground stem

132. Symbol of epitepalous stamens is
(1) $\overbrace{C \quad A}$ (2) $K_{(n)}$
(3) $C_{(n)}$ (4) $\overbrace{P \quad A}$

133. In Australian *Acacia* the petioles modify to form flat, green coloured leaf like structure known as
(1) Cladode
(2) Phylloclade
(3) Phyllode
(4) Bract

134. Select the **odd** one w.r.t. stem modification
(1) Thorn in citrus
(2) Tendril in pumpkin
(3) Pitcher in *Nepenthes*
(4) Tendril in watermelon

135. Perisperm is
(1) Motile male gamete found in angiosperms
(2) Persistent nucellus in the seed
(3) Persistent endosperm in the seed
(4) Found in all seeds

ZOOLOGY

136. Which of the following is **not** a primary metabolite?
(1) Nucleic acid (2) Chlorophyll
(3) Cellulose (4) Albumin

137. Choose the amino acid in which R-group is replaced by $-CH_2OH$?
(1) Glycine (2) Serine
(3) Alanine (4) Tyrosine

138. In which of the given molecules phosphate group is **not** present?
(1) Adenylic acid
(2) Adenosine
(3) Uridylic acid
(4) Guanylic acid

139. Proteins carry out many functions in living organisms **except**
(1) Some of them transport nutrients across cell membrane
(2) Some fight against infections
(3) Some have structural and functional role
(4) They are the primary source of energy in our body

140. Arrange the following molecules in a descending order w.r.t. their average fraction in a cellular mass.
(a) Proteins
(b) Lipids
(c) Nucleic acids
(d) Carbohydrates
(1) (a) – (c) – (d) – (b)
(2) (a) – (d) – (b) – (c)
(3) (c) – (b) – (d) – (a)
(4) (c) – (d) – (b) – (a)

141. Ribulose biphosphate carboxylase oxygenase is the most abundant protein in the
(1) Animal world
(2) Biosphere
(3) Atmosphere
(4) Connective tissue

142. Which nitrogenous compound is present in a phospholipid, lecithin?
(1) Choline (2) Ethanolamine
(3) Serine (4) Inositol

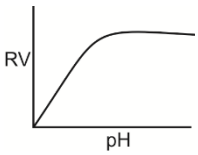
143. Which of the following set is **correctly** matched?
(1) Lysine, thiamine – Amino acids
(2) Starch, glycogen – Storage polysaccharides
(3) Chitin, inulin – Structural polysaccharides
(4) Hyaluronidase, Glucosamine – Homopolysaccharides

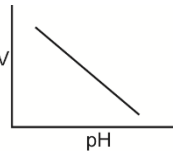
144. Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis are categorized under
(1) Oxidoreductases (2) Hydrolases
(3) Lyases (4) Isomerases

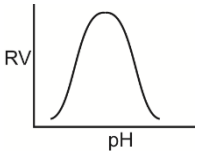
145. In ATP, high energy bonds exist between
(1) Ribose and phosphate
(2) Adenine and ribose
(3) Phosphate and phosphate
(4) Adenine and phosphate

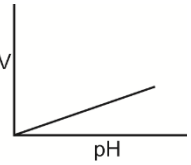
146. Enzymes in our body generally work at
(1) Same pH and temperature optima
(2) Same pH but different temperature optima
(3) Different pH but same temperature optima
(4) Different pH and different temperature optima

147. Which one of the following graph represents the effect of pH on the reaction velocity (RV) of a typical enzyme?

(1) 

(2) 

(3) 

(4) 

148. Lipids differ from carbohydrates in having

	More	Less
(1)	O, H	C
(2)	O	C, H
(3)	C, H	O
(4)	H	C

149. Which of the following is a glycosidic bond?

(1) C – N – C (2) CO = NH

(3) C – H – O (4) C = O

150. Which type of structural organization of protein is present in figure A and B?



	A	B
(1)	Primary	β-Pleated
(2)	α-Helix	Tertiary
(3)	Secondary	Quarternary
(4)	Tertiary	Quarternary

151. The positions at which nitrogen atom is present in guanine are

(1) 1 and 3 only

(2) 2 and 4 only

(3) 1, 3, 7, 9

(4) 1, 3, 6, 9

152. Find the **incorrect** statement.

(1) Prosthetic groups are non protein organic compounds which are firmly attached to the apoenzyme

(2) Apoenzymes are thermostable, catalytic part of enzyme

(3) Catalytic activity of holoenzyme is lost when cofactor is removed from it

(4) Coenzyme take part in transferring small chemical, picking up from one substrate and handing over to another substrate

153. Which metal ion act as cofactor for the proteolytic enzyme, carboxypeptidase?

(1) Mn^{+2} (2) Mg^{+2}

(3) Zn^{+2} (4) Na^{+}

154. Ester bond in nucleic acid involve which two groups of adjacent nucleotides?

(1) OH of sugar and nitrogen of nitrogenous base

(2) Phosphate and hydroxyl group of sugars

(3) Hydroxyl groups of two sugars

(4) Phosphate and hydroxyl group of nitrogenous base

155. A competitive inhibitor of enzyme succinic dehydrogenase is

(1) Malonate

(2) Fumerate

(3) Succinate

(4) Oxamate

156. Choose the effect of presence of non-competitive inhibitor on K_m and V_{max} of enzyme based reaction?

	K_m	V_{max}
(1)	Decline	Same
(2)	Same	Decline
(3)	Increases	Decline
(4)	Decline	Increases

157. Proximal convoluted tubule of nephron in the kidney has ____.

(1) Cuboidal epithelium with cilia

(2) Cuboidal epithelium with microvilli

(3) Simple columnar epithelium

(4) Pseudostratified epithelium with microvilli

158. Mark the **odd** one w.r.t. secretions produced by exocrine glands?

(1) Saliva

(2) Mucus

(3) Ear wax

(4) Insulin

159. Which junctions exist in our skin that prevents the entry of water into body?

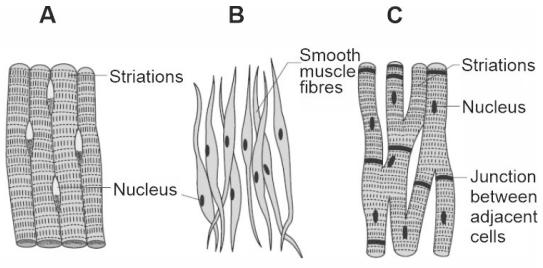
(1) Gap junctions

(2) Tight junctions

(3) Hemidesmosomes

(4) Desmosomes

160. Identify the **correct** sequence of organs where the given muscles marked A, B and C are present



	A	B	C
(1)	Intestine	Stomach	Heart
(2)	Biceps	Stomach	Heart
(3)	Heart	Blood vessel	Biceps
(4)	Tongue	Biceps	Heart

161. Asymmetrically oriented collagen fibres in the dense irregular connective tissue are contributed by

(1) Mast cells

(2) Fibroblasts

(3) Macrophages

(4) Chondrocytes

162. Find the **incorrect** statement.

(1) Neural tissue exerts the greatest control over the body's responsiveness to changing conditions

(2) Neuroglial cells make up more than one-half the volume of neural tissue in our body

(3) Ground substance in blood is synthesized by WBC's

(4) A sheath of tough connective tissue enclose several bundles of muscle fibres

163. Choose the **incorrect** match.

(1)	Bone	–	Hard and non pliable
(2)	Cartilage	–	Resist compression
(3)	Chondrocytes	–	Enclosed in lacunae
(4)	Bone marrow	–	Site of production of all body cells

164. Fusion of how many segments in cockroach form triangular shaped head?

(1) 4 (2) 6

(3) 8 (4) 3

165. Complete the analogy

Hepatic caecae : 6-8 at junction of foregut and midgut :: Malpighian : _____ tubules

(1) 10-15, junction of mesenteron and hindgut

(2) 100-150, junction of midgut and hindgut

(3) 6-8, junction of foregut and hindgut

(4) 80-100, junction of foregut and mesenteron

166. In female cockroach, genital pouch is formed by _____ sternum.

Choose the option which fills the blank **correctly**.

(1) Only 7th, 8th

(2) 7th, 8th and 9th

(3) Only 7th

(4) Only 9th and 10th

167. Spermatophores are

(1) Bundles of sperms – Stored in phallic gland

(2) Asymmetrical phallomeres–Surrounding gonopore

(3) Sperms glued together – Stored in seminal vesicle

(4) Chitinous plates – Helps in holding sperms

168. Different development stages in *Periplaneta* are

(1) Egg – Adult

(2) Egg – Naiad - Adult

(3) Egg – Larva – Pupa – Adult

(4) Egg – Nymph – Adult

169. Uric acid in cockroach is formed in the

(1) Haemolymph

(2) Lumen of gut

(3) Lumen of Malpighian tubules

(4) Epithelium of Malpighian tubules

170. Which one of the following **incorrectly** describes the location of given body parts in the cockroach?

(1) Mushroom gland – 6th – 7th abdominal segments

(2) Ovaries – 2nd – 6th abdominal segments

(3) Testes – 3rd – 6th abdominal segments

(4) Anal cerci – 10th abdominal segment

171. The haemolymph in cockroach involve all the following components **except**

(1) Plasma (2) Haemoglobin

(3) Haemocytes (4) Trehalose

172. Which of the following are paired structures in female cockroach?

(a) Labrum (b) Mandible

(c) Anal cerci (d) Collaterial gland

(e) Oviduct

(1) (a), (b), (c), (d) (2) (b), (c), (e) only

(3) (c), (d), (e) only (4) (b), (c), (d), (e)

173. Choose the option which correctly fills the blanks. Network of trachea in cockroach opens through A of B present on C side of the body.

	A	B	C
(1)	8 pairs	Spiracles	Ventrolateral
(2)	10 pairs	Spiracles	Lateral
(3)	10	Sphincters	Lateral
(4)	8	Sphincters	Dorsolateral

174. Chitinous tergite, pleura and sternite of a particular segment are linked together by

(1) Thin, hard, arthrodial membrane

(2) Thick, hard, articular membrane

(3) Thin, flexible, arthrodial membrane

(4) Thick, solid, sphincters

175. How many moultings does the developmental stage of cockroach undergo to form adult?

(1) 3 (2) 13

(3) 23 (4) 10

176. Which part associated with the gut of cockroach is free of cuticle?

(1) Crop (2) Gizzard

(3) Hepatic caecae (4) Hindgut

177. What is **not** true regarding mesothoracic wings in *Periplaneta*?

(1) Also called tegmina

(2) Opaque, dark and leathery

(3) Helps in flight

(4) Covers the hind wings when at rest

178. Mosaic vision in cockroach have

	Sensitivity	**Resolution**
(1)	More	Less
(2)	Less	More
(3)	High	High
(4)	Low	Low

179. If the supra-oesophageal ganglia of cockroach is destroyed, which activity/activities will be most affected?

(a) Movement of antenna

(b) Vision

(c) Mastication of food

(d) Flattering of wings

(1) (a), (c)

(2) (c), (d)

(3) (a), (b)

(4) (b), (c)

180. Heart of cockroach consists of A chambers and entry of haemolymph in it is associated with B muscles

	A	**B**
(1)	10	Tergosternal
(2)	13	Alary
(3)	12 pairs	Alary
(4)	10 pairs	Abdominal

❑ ❑ ❑